

3 **(a)** Using standard summation formulae, write down an expression in terms of n for $\sum_{r=1}^{2n} r^3$. **[1]**

(b) Hence show that $\sum_{r=n+1}^{2n} r^3 = \frac{1}{4}n^2(an+b)(cn+d)$, where a, b, c and d are integers to be determined. **[5]**